

Soil physicochemical properties and roots promoted preferential flow development after vegetation restoration

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ABSTRACT

Preferential flow, a critical component of soil infiltration, plays a vital role in ecohydrological processes. Large-scale vegetation restoration on the Loess Plateau, China, profoundly alters soil physicochemical properties and root characteristics, thereby influencing the development of preferential flow. However, the long-term effects of vegetation restoration on preferential flow and the underlying mechanisms remain unclear. In this study, double-ring infiltration experiments and dye tracing techniques were employed to compare preferential flow dynamics and their underlying mechanisms under different vegetation restoration types. The transmission mechanisms among soil physicochemical properties, root characteristics, and preferential flow were analysed. The results indicated that preferential flow generally intensified following the conversion of cropland to forestland, shrubland, or grassland, and this intensification increased with increasing years of vegetation restoration. The infiltration rate of preferential flow was 1.87 to 4.88 times greater than that of matrix flow, contributing 61.33% to 76.71% of the total infiltration. Increases in the macroporosity, connectivity, fractal dimension, soil organic matter, root length density, root surface area density, root volume density, and root weight density, increased the soil infiltration capacity and degree of preferential flow, jointly explaining 39.7% of the variance. Root characteristics were the dominant factor influencing preferential flow, independently explaining 23.1% of the variation in preferential flow. These findings enhance the understanding of soil hydrological processes and effects following vegetation restoration on the Loess Plateau and in other arid and semiarid regions.

1. Introduction

Preferential flow (PF) refers to the nonuniform movement of water or solutes through porous media along specific high-conductivity pathways, such as macropores, fractures, or biopores, at a much faster rate than through the surrounding soil matrix. This flow pattern deviates from the uniform seepage described by Darcy's law and is widely present in terrestrial hydrological cycles, significantly influencing soil water infiltration, solute transport, and groundwater recharge (Bouma, 1981; Beven & Germann, 1982). As many as 15 different types of PF have been identified (Lin, 2010). PF can be classified into three main categories on

the basis of its origin and characteristics: macropore flow (Beven & Germann, 1982), finger flow (Glass & Nicholl, 1996), and funnel flow (Kung et al., 2000). Although PF pathways represent only a small fraction of the total soil volume, they can contribute between 11 % and 94 % of the total infiltration (Alaoui, 2015). PF enhances soil infiltration, reduces surface runoff, increases groundwater recharge, and mitigates soil erosion and degradation, thereby playing a crucial role in the terrestrial hydrological cycle (Hu et al., 2019; Wen et al., 2021).

Extensive studies have confirmed that vegetation restoration profoundly alters soil physicochemical properties, particularly soil pore structure, including macroporosity (MP), connectivity (Con), and fractal

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dimension (FD) (Bachmair et al., 2009; Cheng et al., 2018a; Qiu et al., 2023). This can be attributed to two main factors. First, long-term agricultural activities, such as plough, disrupt the original pore network and macroaggregate structure in the topsoil and subsoil (Wu et al., 2023). Second, in abandoned farmland, physical, chemical, and biological interactions among litter layers, root systems, soil, and soil fauna (microorganisms) enhance the soil pore structure (Cui et al., 2022). These changes in soil pore characteristics significantly influence the occurrence and development of PF (Allaire et al., 2009; Sammartino et al., 2015; Zhang et al., 2015; Schwenk et al., 2023). Previous studies have shown that vegetation restoration increases MP and Con, thereby promoting PF development, with varying impacts across different land-use types (Bachmair et al., 2009; Cheng et al., 2018b). Therefore, investigating changes in soil physicochemical properties, particularly pore structure, following vegetation restoration and their effects on PF is highly important. Additionally, since PF occurs throughout the entire soil profile, it is essential to explore how soil properties at different depths influence PF variations after vegetation restoration. In particular, comparing PF characteristics across different vegetation restoration durations within the same land-use type is an important research direction. Plant roots are among the direct factors influencing soil infiltration capacity and the development of PF (Xiao et al., 2024). Research shows that root exudates alter soil hydraulic properties by absorbing large amounts of water, changing the size and stability of aggregates, modifying the soil contact angle, and affecting the viscosity of pore fluids (Zhang et al., 2023). In addition to directly impacting the process of soil infiltration and PF development, roots also indirectly influence soil hydraulic properties through root-induced changes in soil properties, such as bulk density (BD), the proportion of water-stable aggregates > 0.25 mm (AGC), and MP. Studies have shown that in alfalfa communities, 60 % of flow paths are associated with root channels or decayed roots (Stokes et al., 2009; Guo et al., 2019). Moreover, roots can increase soil surface roughness and create continuous macropore spaces, thereby promoting the development of PF, increasing infiltration capacity, and reducing surface runoff (Bodner et al., 2014). The root system characteristics of different vegetation types significantly influence soil water movement and PF processes (Wang et al., 2024a,b). Archer et al. (2002) reported that shrubs and trees with taproot systems transport soil water to deeper layers more effectively than herbaceous plants with fibrous roots do, thereby promoting both vertical and lateral development of PF. Cai et al. (2024) indicated that PF increased with forest age in artificial black locust forests. However, further research is needed to elucidate the mechanisms by which plant root systems alter soil pore structure and subsequently affect PF dynamics.

In recent years, research on the quantification of PF across various scales and methods has gradually advanced. The current techniques for studying PF primarily include tracer methods, breakthrough curve analysis, tomography, soil moisture dynamic monitoring, and ground-penetrating radar (Allaire et al., 2009). The dye tracer method is the most commonly used approach for studying PF, as it provides a direct visual representation of the flow paths and patterns. When combined with subsequent image processing and analysis, this method enables PF dynamics to be described quantitatively (Jiang et al., 2017). Additionally, the dye tracer method is time-efficient and cost-effective. As a result, it is used in large variety applications worldwide. Although many studies have used dye tracer method to investigate the effects of different land use types on PF in the Loess Plateau (LP), the research areas have generally been small, leading to inconsistent conclusions. For example, Qiu et al. (2023) reported that the maximum dye depth (MDD) of PF in shrubland was greater than that in forestland and grassland. However, Mei et al. (2018) found that forestland exhibits a greater MDD. These plot-scale studies have deepened our understanding of the effects of local land use types on PF. However, due to the limited scope of these independent case studies, their results are difficult to generalize to the entire LP. The question of which land use type exhibits the most pronounced PF development after vegetation restoration remains debated.

To extrapolate plot-scale PF data to a larger regional scale, we conducted a literature review to explore the universal law of PF under different land use types. With technological advancements, X-ray computed tomography (CT) has emerged as a vital tool in PF research because of its high-resolution and nondestructive imaging capabilities (Petrovic et al., 1982). CT scanning enables the acquisition of three-dimensional soil pore structures and their spatial distributions without disturbing soil samples, providing more precise data and insights into the relationship between soil pore architecture and PF (Tobella et al., 2014; Wei et al., 2024). Although CT scanning technology is expensive and complex, its use is mainstream in PF studies due to its nondestructive nature, accuracy, and high resolution (Udawatta et al., 2008). However, studies using high-precision industrial CT scanning to quantify the microscopic pore structure of soils under different vegetation restoration types in the LP and to reveal its impact on PF mechanisms remain relatively limited. A double-ring infiltrometer is the standard method for measuring soil surface infiltration. It can measure both the infiltration process and the infiltration rate of soil. Zhang et al. (2017a, 2019) proposed a field experimental method for measuring PF infiltration with two double-ring infiltrometers, where one ring was covered with 3 cm of fine sand to block the PF path. By measuring the difference in total infiltration and matrix infiltration in adjacent areas, the PF infiltration rate and its contribution could be determined. Compared with other methods, this approach is relatively cost-effective, simple, and easy to perform; therefore, its use can provide new insights for quantifying PF (Tao et al., 2019). Furthermore, Zhao et al. (2023) conducted a quantitative study on the cover material. By filling the outer ring of the infiltrometer with different proportions of soil-sand mixtures to shield the PF path, they measured the matrix flow (MF) infiltration process. This method not only allows for more accurate quantification of MF, but also effectively distinguishes PF and MF in different soil types, overcoming the issue of insufficient accuracy in traditional double-ring methods for high-porosity soils. However, how to use an improved double-ring infiltrometer to quantify the contribution of PF to total infiltration under different vegetation restoration types in the LP remains an urgent issue to be addressed.

The Chinese LP is one of the regions that is most severely affected by soil erosion worldwide. In recent decades, to combat severe soil erosion in the region, China has implemented large-scale soil and water conservation and ecological engineering projects on the LP (Gao et al., 2011). Measures such as the Grain for Green Program and afforestation have significantly increased vegetation cover on the LP, with the vegetation index increasing from 31.6 % in 1999, when the Grain for Green Program was initiated, to 67 % in 2020 (Pan et al., 2024). Large-scale vegetation restoration has significantly altered land use and cover patterns, leading to substantial changes in soil pore structure and root system characteristics, which ultimately influence PF and have profound impacts on ecohydrological processes (Chen et al., 2021). Understanding the changes in PF driven by vegetation restoration and their underlying mechanisms is crucial for soil hydrology effect in the LP. However, while numerous studies have compared the effects of different land-use types on PF, research on the influence of vegetation restoration duration and soil depth remains limited. Moreover, studies examining the transmission mechanisms from root characteristics to soil physicochemical properties and subsequently to PF are even scarcer.

To gain a deeper understanding of the characteristics and mechanisms of PF on the Chinese LP, we conducted experiments combining double-ring infiltrometers with dye tracer techniques. These experiments measured soil physical and chemical properties, macropore structure, and plant root systems. The objectives of this study are threefold. (1) The differences of PF under different vegetation restoration types and restoration durations were clarified, and the characteristics of PF of different land use types in the LP were analyzed by meta-analysis. (2) The contributions of PF and matrix flow to infiltration across different land-use types were quantified. (3) The critical factors influencing the PF were identified, and their impact levels were

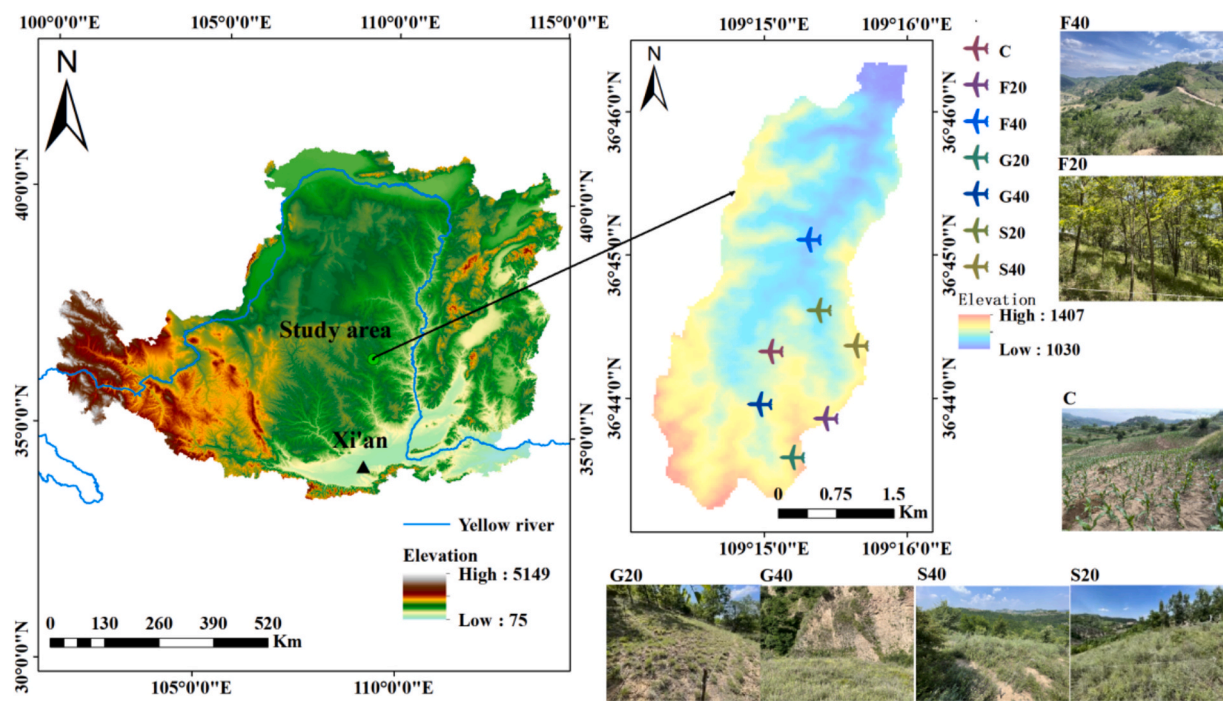


Fig. 1. Information and photos of the study area and land-use types.

Table 1
List of abbreviations in this paper.

Abbreviations	Full name	Abbreviations	Full name
Soil properties		Preferential flow index	
BD	Bulk density	PF	Preferential flow
SOM	Soil organic matter	MF	Matrix flow
IMC	Initial moisture content	MDD	Maximum dye depth
AGC	Proportion of water-stable aggregates ≥ 0.25 mm	DC	Dye coverage
MP	Macroporosity	MFD	Matrix flow depth
Con	Connectivity	PF _f	Preferential flow fraction
FD	Fractal dimension	LI	Length index
Root	Root length density	PF-CI	Preferential flow composite index
		SPW	Stained path width
RSD	Root surface area density	Infiltration index	
RVD	Root volume density	IIR	Initial infiltration rate
RWD	Root weight density	SIR	Stable infiltration rate
Statistical method	Principal component analysis	IIR-P	Initial infiltration rate of preferential flow
		IIR-M	Initial infiltration rate of matrix flow
PCA	Principal component analysis	SIR-P	Stable infiltration rate of preferential flow
RDA	Redundancy analysis	SIR-M	Stable infiltration rate of matrix flow
VPA	Variance partitioning analysis	K _s	Saturated hydraulic conductivity
PLS-PM	Partial least squares path modeling		

assessed. This research provides scientific evidence for a more profound understanding of the hydrological effects of soil under vegetation restoration on the LP during this new era.

2. Materials and methods

2.1. Study areas

The study area is in the Zhifanggou watershed in northern Shaanxi Province, China (36°42'–36°46'N, 109°13'–109°16'E) (Fig. 1). This watershed is part of the typical loess hilly landscape of the Chinese LP and is positioned in the transition zone between warm temperate forests and steppes. Spanning an area of approximately 8.27 km², the elevation varies from 1032 to 1393 m above sea level. The average annual precipitation is 458.30 mm (1990–2023) (Fig. S1d), with significant seasonal variation. Most precipitation occurs as heavy rainfall during the period from June to September, accounting for approximately 72.8 % of the total annual precipitation (Fig. S1a). The region has an average annual temperature of 8.8 °C (Fig. S1c). The soil type is silty loam, which is characterized by a loose, uniform texture and pale yellow colour. Owing to low surface vegetation cover, the soil is highly vulnerable to erosion. The average annual runoff is 14.95 mm, and the sediment yield is 3.18 × 10⁴ tons (1993–2019) (Fig. S1b). Since 1973, various soil and water conservation measures have been introduced, resulting in significant vegetation recovery. The primary land-use types include forestland, shrubland, grassland, and farmland, as well as smaller areas of orchards and abandoned land (Wang et al., 2014; Qiu et al., 2022).

2.2. Experimental design and process

A space-for-time substitution approach was used to select forestland, grassland, and shrubland with 20 and 40 years of vegetation restoration in the study area, with farmland serving as the control (Lovell et al., 2023). Three plots (15 m × 20 m) were selected as replicates for each land use type, resulting in a total of 21 plots. During plot selection, site conditions such as slope, slope aspect, slope position, and elevation were fully considered to ensure that all plots were originally farmland before conversion, thus ensuring good comparability among plots. Detailed

Table 2
Basic information of the sampling plots.

Land-use	Vegetation	Longitude	Latitude	Elevation (m)	Aspect	Slope (°)	Position	Coverage (%)	Year
Forestland	RP	109.263	36.7309	1273.18	north-facing	21.67(1.56)	middle	50(4.67)	20
Forestland	RP	109.2552	36.7517	1105.73	north-facing	23.67(4.22)	middle	61.33(4.22)	40
Shrubland	CK, SD	109.2564	36.7394	1171.38	north-facing	19.33(2.89)	middle	52.67(2.89)	20
Shrubland	CK, SD	109.2607	36.7394	1263.03	north-facing	23.33(1.56)	middle	60.67(8.22)	40
Grassland	AG, AC, BI, AA, SB	109.2533	36.7264	1337.34	north-facing	17.33(4.22)	middle	44.67(4.22)	20
Grassland	AG, AC, BI, AA, SB	109.2495	36.7326	1219.92	north-facing	24.33(1.56)	middle	60.33(2.89)	40
Cropland	Corn	109.2508	36.7387	1214.05	north-facing	5.33(1.56)	middle	16.67(1.56)	—

The values outside the brackets represent the mean, while the values inside the brackets indicate the standard deviation; RP: *Robinia pseudoacacia*; CK: *Caragana korshinskii*; SD: *Sophora davidii*; AG: *Artemisia gmelinii*; AC: *Artemisia capillaris*; BI: *Bothriochloa ischaemum*; AA: *Astragalus adsurgens*; SB: *Stipa bungean*.

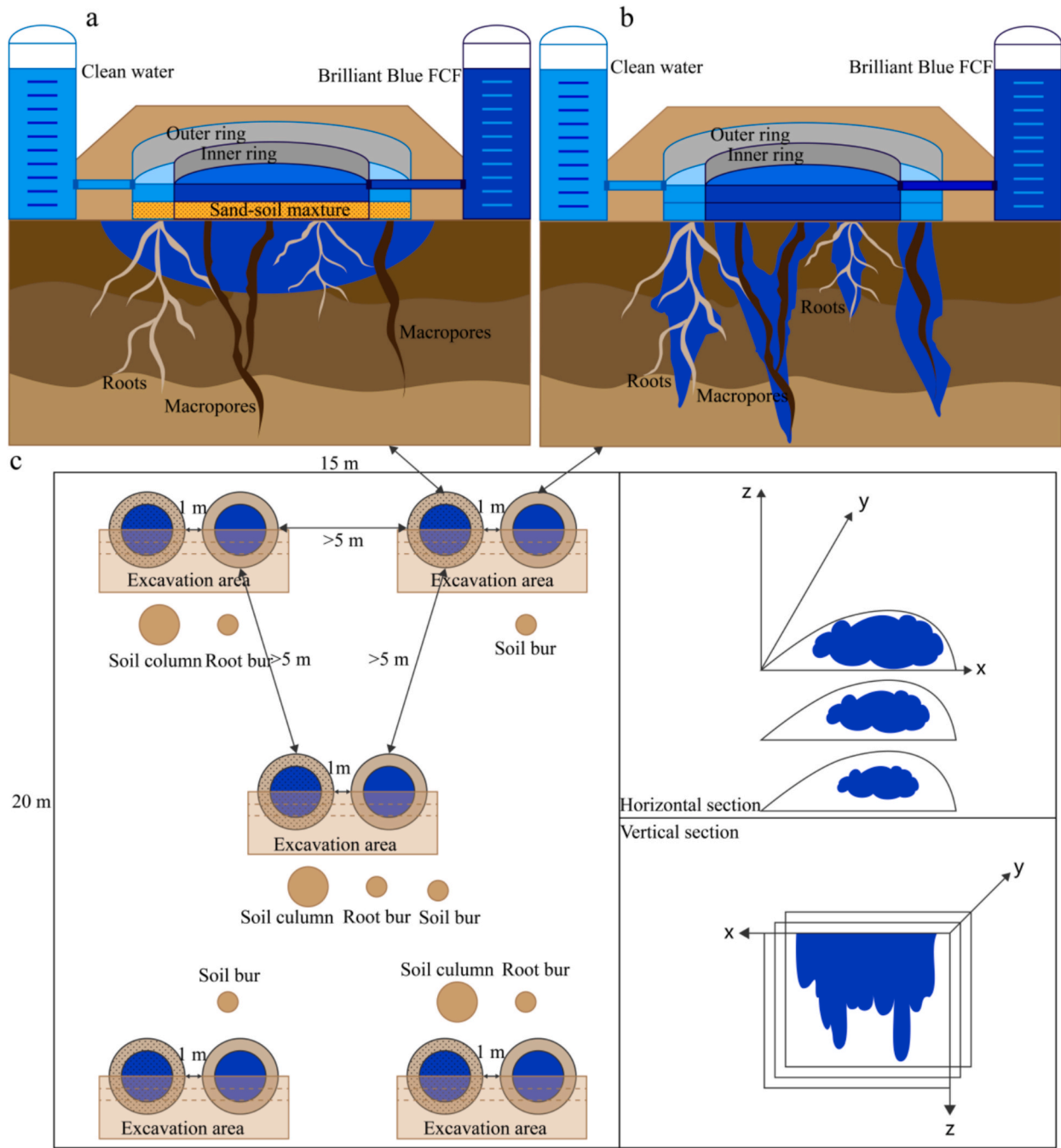


Fig. 2. Schematic diagram of the double-ring infiltration and test sample point layout. a is used to measure the soil matrix infiltration process, b is used to measure the total infiltration process, and c illustrates the setup of the double-ring infiltration and dye tracing experiments, as well as the excavation schematic of horizontal and vertical stained profiles.

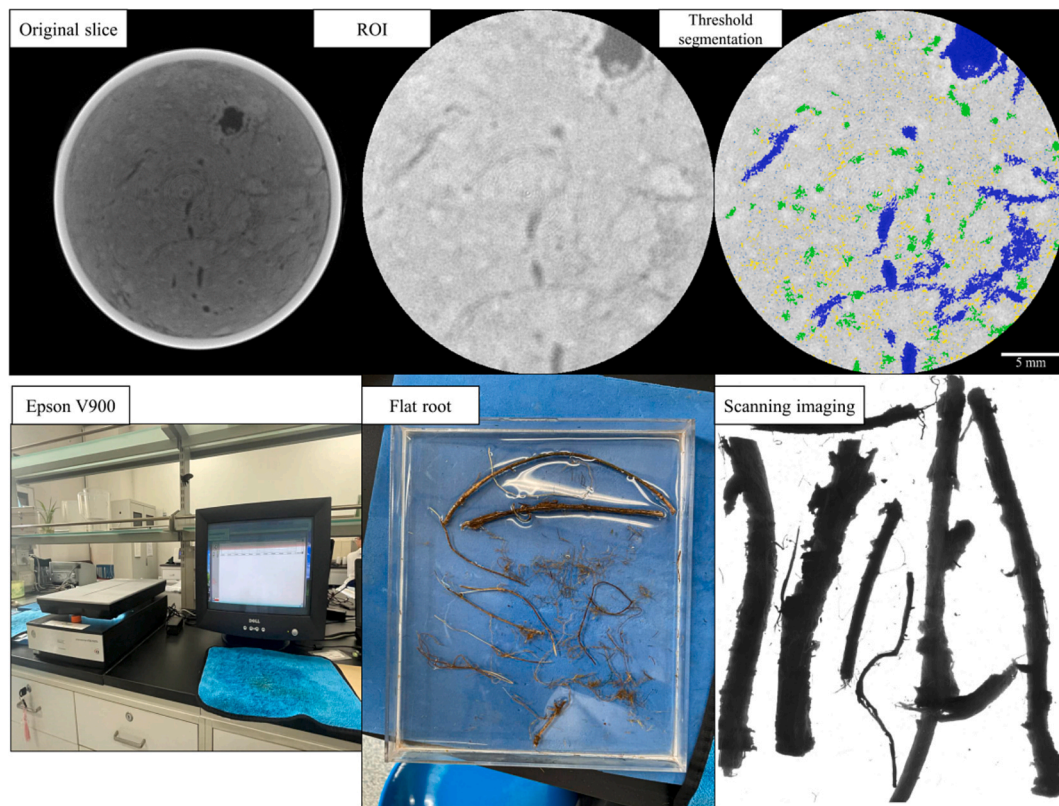


Fig. 3. Flow chart of soil pore structure CT scanning and plant root scanning.

information about the plots is provided in Table 2. Notably, there was no precipitation in these plots for at least two weeks prior to the sampling and infiltration experiments. The field experiments were conducted from May to June 2023.

In each plot, five sampling points were established using the diagonal method, ensuring a minimum distance of 5 m between each point (Fig. 2c). Prior to the infiltration experiment, the surface litter (0–5 cm) was removed, and the soil was levelled. At each sampling point, two double-ring infiltrometers, with inner and outer diameters of 30 cm and 45 cm, respectively, were inserted 10 cm into the soil to block the lateral PF pathway (Qiu et al., 2023). The infiltrometers were supplied with water from two 10-litre Mariotte bottles, and the constant head method was employed to measure infiltration parameters. A right-sided double-ring infiltrometer was used to measure undisturbed in situ soil infiltration (Fig. 2b), whereas a left-sided infiltrometer was used to assess soil matrix infiltration (Fig. 2a). Filter papers 5 cm in diameter were placed in the inner and outer rings to regulate water flow and prevent disruption of the soil surface structure. During the undisturbed in situ infiltration experiment on the right side, a Brilliant Blue FCF dye solution (4.0 g/L; Flury & Flühler, 1995) was added to the inner ring, and clear water was introduced to the outer ring until the preset water level was reached (the ponding depth was 4 cm). The water level was maintained, and the Mariotte bottle readings were recorded at various intervals until the infiltration rate in the inner ring stabilized, indicating the end of the experiment. The volume of infiltrated water was recorded at 15 time intervals. Previous studies have shown that the surface soil on the LP reaches a stable infiltration rate (SIR, the list of abbreviations is shown in Table 1) within 20–30 min. To control the total infiltration duration, the total infiltration time for all plots was set to 40 min (Gu et al., 2019). On the basis of continuous measurements, the average infiltration rates for the first 3 min and the last 10 min were calculated and designated the initial infiltration rate (IIR, 10 °C) and the SIR (10 °C), respectively (Qiu et al., 2023). To better assess the infiltration capacity under the different flow modes, we calculated the initial infiltration rate of MF (IIR-M),

initial infiltration rate of PF (IIR-P), stable infiltration rate of MF (SIR-M), and stable infiltration rate of PF (SIR-P) indices for each land-use type.

Perroux and White (1988) reported that a sand–soil mixture, as a cover material, has a slightly higher K_s than does the soil matrix, which effectively blocks PF. The higher the soil proportion is, the greater the shielding effect on PF. To measure the K_s of the soil matrix, disturbed soil samples were collected and processed through grinding, sieving, and other treatments to obtain the K_s of the surface soil matrix (0–10 cm) in different plots. The K_s values of the surface soils from the cropland, G20, G40, S20, S40, F20, and F40 plots were 0.43 cm/min, 0.64 cm/min, 0.69 cm/min, 0.67 cm/min, 0.77 cm/min, 0.68 cm/min, and 0.73 cm/min, respectively. The K_s values of the sand–soil mixtures with different ratios (1:1, 1:2, 2:3, 1:3, 1:4, and 1:5) were 2.15 cm/min, 1.37 cm/min, 0.84 cm/min, 0.75 cm/min, 0.44 cm/min, and 0.27 cm/min, respectively. Finally, for the cropland, a 1:4 sand–soil mixture was selected; for G20, G40, S20, F20, and F40, a 1:3 sand–soil mixture was selected; and for S40, a 2:3 sand–soil mixture was selected as the cover material. For the left-side soil matrix infiltration experiment, a layer of gauze was first laid on the surface inside the ring, followed by a 3-cm thick sand–soil mixture (Zhao et al., 2023). The subsequent measurement process was the same as that in the right-side double-ring experiment. After the experiment, the infiltration amount and rate of PF were calculated to analyse the quantitative relationship between PF and MF. Twenty-four hours after infiltration ended, a vertical profile with a depth of 1 m was excavated along the inner half of the ring. After each vertical profile was excavated, the depth of the profile was determined with a ruler and marked, and high-definition digital images of the profile were taken with a digital camera (Nikon D7500, Japan).

$$i_{MF} = i_{Left} \quad (1)$$

$$i_{PF} = i_{Right} - i_{Left} \quad (2)$$

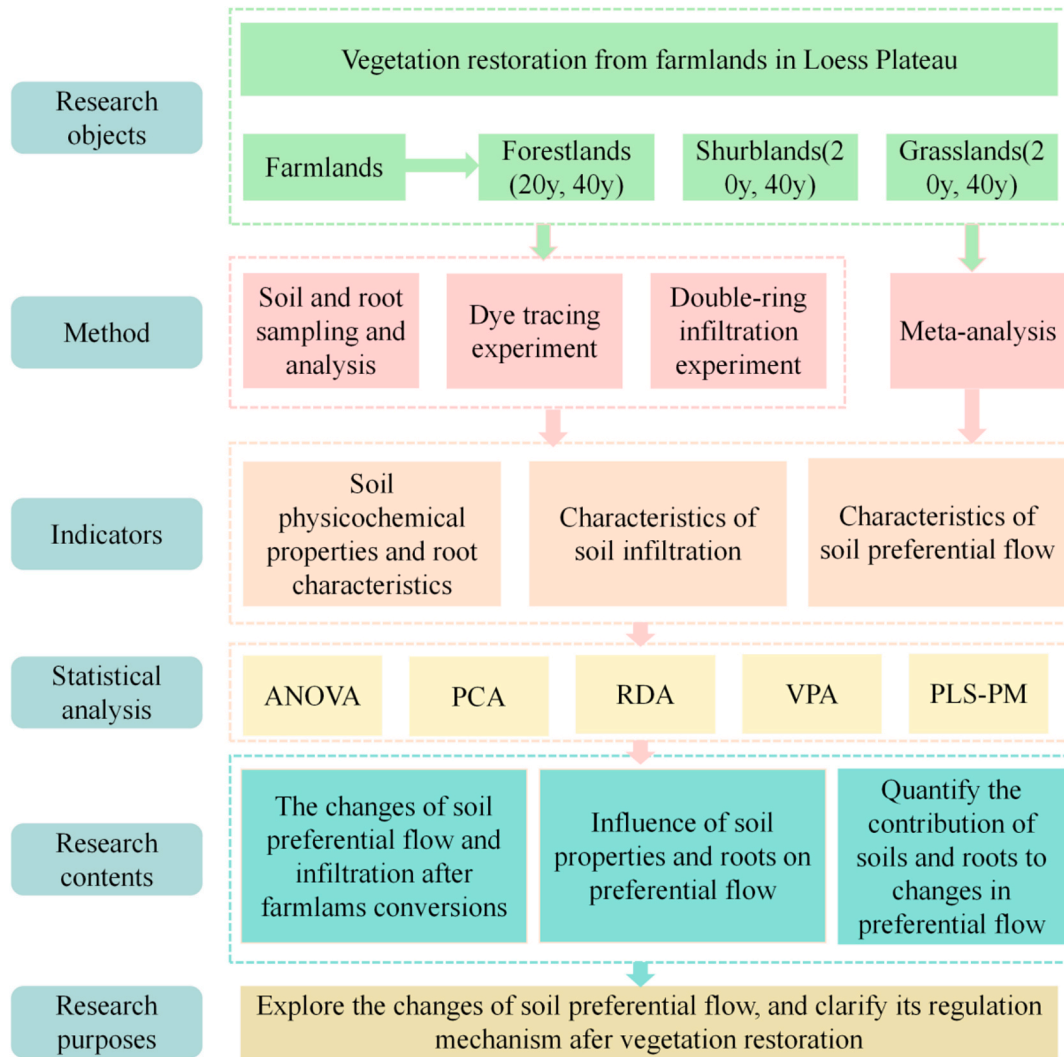


Fig. 4. The analytical framework of this paper. ANOVA: Analysis of Variance; PCA: Principal component analysis; RDA: Redundancy analysis; VPA: Variance partitioning analysis; PLS-PM: Partial least squares path modeling.

where i_{MF} represents the matrix flow infiltration rate (mm/min), i_{Left} represents the left-side double-ring infiltration rate (mm/min), i_{PF} represents the PF infiltration rate (mm/min), and i_{Right} represents the right-side double-ring infiltration rate (mm/min).

2.3. Collection and measurement of soil and roots

At each experimental site, disturbed soil samples were collected on the opposite side of the vertical dye profile using 100 cm³ ring knives and 2000 cm³ stainless steel boxes at 20 cm depth intervals (7 land-use types × 5 plots × 3 soil layers × 2 replicates). These samples were analysed for BD, particle composition, AGC, soil organic matter (SOM), and initial moisture content (IMC). Undisturbed soil samples were also collected using 100 cm³ ring knives for subsequent macropore analysis. Near each dye tracing point, root samples were collected from the undisturbed area using a 9 cm diameter root auger at a depth of 20 cm. Soil from the same layer was combined into a composite sample, resulting in 105 root samples (7 land-use types × 5 plots × 3 soil layers).

BD and the IMC were determined using the oven-drying method (Bao, 2007). The SOM was measured with the dichromate oxidation method (Dong et al., 2022). The pH of the soil was recorded using a pH meter with a soil-to-water ratio of 2.5:1. The particle size distribution was analysed with a Mastersizer 2000 laser particle sizer (Malvern Instruments Ltd., Malvern, UK), which classifies particles according to the

USDA soil taxonomy into sand (2.0–0.05 mm), silt (0.05–0.002 mm), and clay (< 0.002 mm). AGC was measured using the wet sieving method (Kemper & Rosenau, 1986). The soil and root samples were collected with a root auger, and the soil was shaken to remove loose particles before the remaining soil containing the roots was stored in zip-lock bags. In the laboratory, the roots were separated using a 0.25 mm sieve, rinsed with water, and dried with nonwoven fabric. The roots were then scanned at 300 dpi with an Epson V900 photo scanner (Japan) (Fig. 3). The root length density (RLD, cm/100 cm³), root surface area density (RSD, cm²/100 cm³), and root volume density (RVD, cm³/100 cm³) were quantified using the WinRHIZO root analysis system (2009a, Regent Instruments, Quebec, QC, Canada). Finally, the roots were oven-dried at 60 °C for 72 h and weighed to the nearest 0.01 mg, after which the root weight density (RWD, mg/100 cm³) was calculated (Fig. 3) (Wang et al., 2024a,b). The formulas are as follows:

$$RLD = RL/V \quad (3)$$

$$RSD = RS/V \quad (4)$$

$$RVD = RV/V \quad (5)$$

$$RWD = RW/V \quad (6)$$

where: RL is the total root length (cm), RS is the total root surface area

Table 3Soil physicochemical properties at the different depths and land-uses (Means $\pm$ SEss).

Soil depth (cm)	Land-use	BD (g/cm ³)	SOM (g/kg)	pH	Clay (%)	Silt (%)	Sand (%)	AGC (%)	MP (%)	Con	FD
0–20	F20	1.2(0.05) aB	6.09(1.58) bcA	8.35 (0.08)aB	<u>16.00</u> (1.63)aA	59.26 (0.49)cA	<u>24.74</u> (2.11)abA	70.49(8.28) abA	<u>11.62</u> (0.52)bA	<u>3468.33</u> (384.53)cA	2.49 (0.05)bA
	F40	1.22(0.03) aB	<u>9.12</u> (3.95) abcA	8.32 (0.21)aA	13.21 (0.88)bB	61.25 (1.66)bcB	<u>25.54</u> (2.50)abA	75.81 (4.03) aA	<u>14.39</u> (0.70)aA	10488.67 (1490.20)aA	2.61 (0.03)aA
	S20	1.25(0.03) aB	<u>9.90</u> (7.97) abA	8.26 (0.06)aB	17.59 (0.61)aA	59.80 (1.10)cB	<u>22.61</u> (1.64)abA	<u>57.22</u> (4.22) cA	<u>11.75</u> (0.47)bA	2806(482.36) bA	<u>2.37</u> (0.03)cA
	S40	1.27(0.06) aA	<u>15.37</u> (2.81)aA	8.25 (0.18)aA	16.51 (0.55)aA	61.82 (1.97)bcB	<u>21.67</u> (1.80)bcA	<u>58.80</u> (3.85) bcA	15.19 (0.74)aA	<u>6059.33</u> (324.16)bcA	2.37 (0.09)cA
	G20	1.19(0.07) aB	<u>4.52</u> (1.09) cA	8.42 (0.05)aB	<u>13.44</u> (2.03)bA	59.89 (0.38)cA	26.67 (1.80)aA	<u>39.33</u> (7.05) dA	<u>11.45</u> (0.83)bA	<u>2732.33</u> (216.61)cA	<u>2.21</u> (0.01)dA
	G40	1.21(0.08) aA	15.83 (2.16)abcA	8.34 (0.02)aB	16.80 (0.48)aA	65.03 (1.71)aA	<u>18.17</u> (1.70)cA	<u>53.75</u> (0.88) cA	<u>14.06</u> (0.58)aA	<u>5971.67</u> (505.06)bA	<u>2.51</u> (0.05)abA
	C	1.32 (0.03) aB	<u>7.33</u> (1.11) bcA	8.25 (0.07)aA	10.05 (0.93)cA	63.25 (1.17)abA	<u>26.25</u> (1.18)aA	<u>31.04</u> (7.26) dA	<u>7.83</u> (0.18)cA	<u>2451</u> (226.45)cA	2.04 (0.01)eB
	F20	<u>1.31</u> (0.03) abA	4.54(0.44) cdB	8.54 (0.02)aA	14.59 (0.70)bcdA	60.67 (1.69)bA	<u>24.74</u> (1.99)aA	59.88(5.78) aAB	9.12 (0.42)aB	2870(327.20) bAB	<u>2.53</u> (0.09)aA
	F40	1.30(0.06) abAB	9.05(1.86) bA	8.44 (0.06)bcA	16.20 (1.75)abA	64.81 (0.60)aA	18.99 (1.35)bB	60.16 (2.64) aAB	10.27 (0.60)aB	4792.67 (1093.65)aB	2.63 (0.04)aA
	S20	1.35(0.08) aAB	4.9(1.95) bcA	8.42 (0.04)cA	18.38 (1.55)aA	<u>64.22</u> (1.57)aA	17.40 (1.80)bB	41.00(18.28) abcAB	9.01 (0.87)aB	<u>3080</u> (160.89) bA	2.22 (0.03)cB
Soil depth (cm)	Land-use	BD (g/cm ³)	SOM (g/kg)	pH	Clay (%)	Silt (%)	Sand (%)	AGC (%)	MP (%)	Con	FD
20–40	S40	<u>1.30</u> (0.04) abA	6.43(0.76) bcB	8.42(0.05) cA	15.37 (0.13)bcA	65.86 (0.46)aA	18.77 (0.38)bB	43.50(8.44) abAB	10.37 (0.68)aB	3100.33 (459.59)bB	2.38(0.05) bA
	G20	1.31(0.03) abA	3.8(0.23) dB	8.52(0.02) abA	13.23 (0.78)cdA	<u>61.07</u> (0.47)bA	25.70 (0.46)aA	30.39(0.83) bcA	10.07 (0.36)aAB	2294.67 (145.70)bcB	2.18(0.02) cdA
	G40	1.21(0.08) bA	11.52 (0.41)aA	8.36(0.01) cdB	16.80 (0.48)abA	65.04 (1.71)aA	18.16 (1.70)bA	52.87(8.52) aA	10.24 (0.82)aB	3579.33 (854.26)abB	2.19(0.05) cdB
	C	1.42 (0.04) aA	5.61(0.97) bcdA	8.30(0.05) dA	<u>12.38</u> (0.88)dA	63.58 (0.28)abA	23.99 (0.98)aA	23.00(4.91) cAB	5.94(0.46) bB	1268(175.52) cB	<u>2.10</u> (0.02) dA
	F20	<u>1.31</u> (0.03) cdA	3.97(0.65) bcB	<u>8.56</u> (0.02) aA	14.56 (1.41)bcA	<u>61.36</u> (2.17)bcA	24.08 (2.71)aA	50.92 (7.81) aB	8.33(0.46) abB	2385.33 (182.77)bcB	2.34(0.07) bcA
	F40	<u>1.41</u> (0.07) abcA	7.07(0.80) aA	<u>8.46</u> (0.07) abA	<u>19.00</u> (0.49)aA	<u>64.89</u> (1.03)aA	16.11 (1.49)bB	49.36 (10.56)aB	8.88(1.24) abB	3834.33 (1046.80)aB	2.51 (0.04)aB
	S20	<u>1.49</u> (0.06) aA	4.81(0.20) abA	<u>8.48</u> (0.04) abA	<u>19.19</u> (0.29)aA	63.79 (1.21)abA	17.02 (1.34)bB	29.76(4.60) bB	7.56(0.82) bB	2390(183.49) bcA	2.10(0.03) efC
	S40	1.29(0.07) dA	7.31 (1.38) aB	<u>8.47</u> (0.03) abA	<u>16.77</u> (0.85)abA	67.16 (0.11)aA	16.07 (0.75)bB	32.34(9.73) bB	8.91(0.58) abB	3396(182.65) aB	<u>2.43</u> (0.07) abA
	G20	<u>1.35</u> (0.01) bcdA	3.49(0.34) cB	8.58 (0.02)aA	13.32 (0.66)cdA	60.44 (1.26)cA	26.24 (1.59)aA	16.76(1.14) bcB	9.35(0.67) aB	2014.33(40.84) cB	2.19(0.04) deA
	G40	<u>1.26</u> (0.05) dA	6.34(0.96) aB	<u>8.51</u> (0.00) abA	<u>17.96</u> (2.31)aA	<u>66.56</u> (1.14)aA	15.48 (3.43)bA	31.07(9.95) bB	9.92 (0.45) aB	3020(259.26) abB	2.24(0.08) cdB
	C	<u>1.43</u> (0.04) abA	4.65(0.66) aA	<u>8.40</u> (0.11) bA	11.51 (1.13)dA	<u>64.27</u> (0.38)abA	24.22 (1.48)aA	10.97(3.30) cB	5.16(0.12) cC	1311.33 (216.86)dB	2.04(0.01) fB
Soil depth (cm)	Land-use	BD (g/cm ³)	SOM (g/kg)	pH	Clay (%)	Silt (%)	Sand (%)	AGC (%)	MP (%)	Con	FD
P value	Depth	<0.01	<0.01	<0.01	<0.01	0.018	<0.01	<0.01	<0.01	<0.01	<0.01
P value	Land-use	<0.01	<0.01	<0.01	0.002	<0.01	<0.01	<0.01	<0.01	<0.01	<0.01
	D $\times$ L	0.373	0.027	0.959	0.032	0.083	0.04	0.968	0.003	<0.01	<0.01

Soil texture classification was conducted based on the USDA. The values outside the brackets represent the mean, while the values inside the brackets indicate the standard deviation. The bold entries indicate the maximum values among the four land-use categories, while the underlined values represent the highest values for each of the three soil depths. Significant differences in depth are indicated by different uppercase letters within the same column for a specific land-use type ($P < 0.05$). Significant differences between land-use types at the same depth are marked by different lowercase letters within the same column ($P < 0.05$) (Duncan's test). The abbreviations used are as follows: C stands for cropland; G20 and G40 represent grasslands restored for 20 and 40 years, respectively; S20 and S40 correspond to shrublands with 20 and 40 years of vegetation restoration; and F20 and F40 represent forestlands restored for 20 and 40 years respectively.

(cm²), RV is the total root volume (cm³), RW is the root dry weight (mg), and V is the soil volume from which roots were extracted for each layer (2000 cm³).

The soil cores were scanned with a nanoVoxel1280 industrial CT scanner (Sanying Precision Instruments Co., Ltd., China) at a peak voltage of 60 kV and a current of 0.6 mA, with a pixel matrix of 1280 pixels $\times$ 1280 pixels and an imaging area of 128 mm $\times$ 128 mm. The high-frequency jitter algorithm function was employed, providing a resolution of 0.047 mm. We defined macropores as those with a diameter greater than 0.047 mm. The slice images were analysed with Avizo3D (version 2020). First, a region of interest with a diameter of 30 mm was selected from the centre of the slice images to eliminate interference caused by the edges of the ring knife. The "Interactive thresholding"

command was subsequently used to identify the macropores, and calculations were performed to determine the MP, Con, and FD (Houston et al., 2017) (Fig. 3).

2.4. Image analysis

The soil profile images were processed with the Adobe Photoshop CC 2022 and Image ProPlus v6.0 software packages. First, spatial correction of the images was performed with the lens correction function in Adobe Photoshop CC 2022, with the preset ruler at the edge of the soil profile as the reference plane for image spatial correction (Wang et al., 2020). Next, the images were binarized with the colour replacement function in Adobe Photoshop CC 2022. A tolerance level was set to replace the

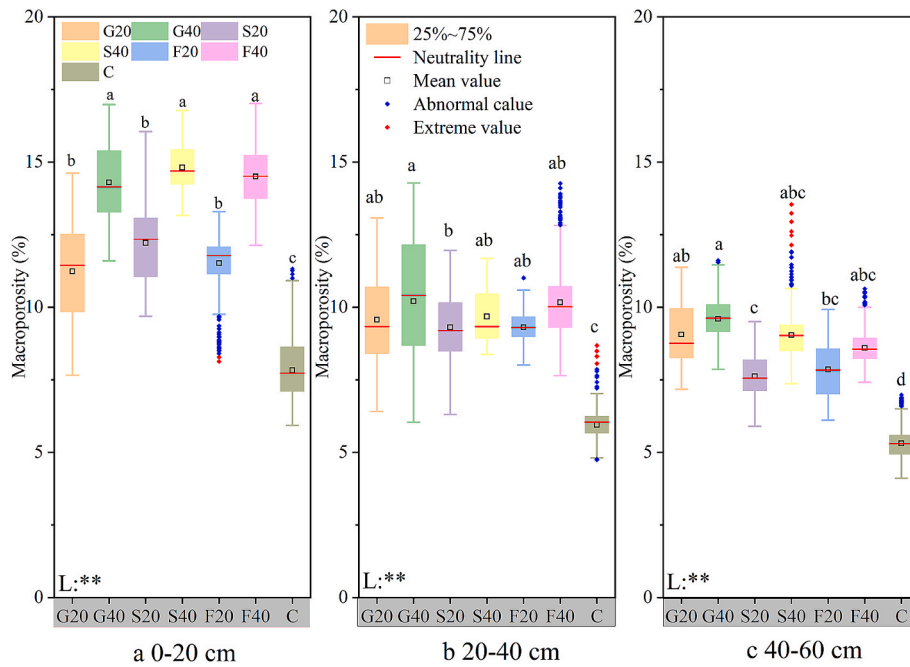


Fig. 5. Panel a shows the change in macropores in the 0–20 cm, 20–40 cm, and 40–60 cm layer. Lowercase letters in the figure indicate significant differences between land-use types at the $P = 0.01$ level.

stained areas with black (0) and the unstained areas with white (255), converting the images into binary matrix images composed only of grayscale values of 0 and 255; the images were then saved in TIFF format. The TIFF images were then imported into Image ProPlus v6.0 software, where the Bitmap Analysis function in Measure was used to binarize the images and output a binary matrix of 500×1000 made up of values of 0 and 255. These data were then stored in Microsoft Excel files for further analysis of the stained images (Jiang et al., 2017).

2.5. Calculation of the PF index

The maximum dye depth (MDD) refers to the deepest extent of dye penetration in the soil profile (cm).

The dye coverage (DC, %) refers to the proportion of the total stained area to the total area of the soil profile (Flury & Flühler, 1995). The formula for calculating DC is as follows:

$$DC = \left(\frac{D}{D + ND} \right) \times 100\% \quad (7)$$

where D (cm^2) and ND (cm^2) represent the stained area and the unstained area, respectively. In this study, the total width of the soil profile was 300 mm (300 pixels).

The matrix flow depth (MFD) refers to the critical soil depth between the matrix flow zone and the PF zone in the stained soil profile, measured in centimetres. In this study, matrix flow depth is defined as the soil depth corresponding to a dye coverage greater than 80 % (Van Schaik, 2009).

The PF fraction (PF_f) is the proportion of the PF area to the total stained area in the soil profile (Van Schaik, 2009). The formula is as follows:

$$PF_f = \left(1 - \frac{MFD \times W}{TSA} \right) \times 100\% \quad (8)$$

where W represents the width of the soil vertical profile (cm) and TSA represents the total stained area of the soil vertical profile (cm^2). The greater the PF fraction is, the greater the degree of PF development.

The length index (LI , %) is the sum of the absolute values of the differences in dye coverage between consecutive layers in the vertical

profile (Tobella et al., 2014). The formula for calculating the LI is as follows:

$$LI = \sum_{i=1}^n |DC_{i+1} - DC_i| \quad (9)$$

where i represents the soil layer in the vertical profile, DC_i and DC_{i+1} represent the dye coverage (%) of the i -th and $(i + 1)$ -th soil layers, respectively and n represents the total number of soil layers in the profile. In this study, the vertical profile was divided into 1 mm soil layer intervals.

The PF composite index (PF-CI) is a composite variable derived from the first principal component axis of principal component analysis (PCA), which is based on the MDD, MFD, PF_f , DC, and LI (Gao et al., 2014; Yu et al., 2023).

The stained path width (SPW, mm) refers to the width of individual stained paths at a specific soil layer depth (Weiler & Flühler, 2004). The formula for calculating the SPW is as follows:

$$SPW = \frac{SN}{R} \quad (10)$$

where SN represents the number of pixels occupied by individual stained paths, and R is the image resolution, which in this study is $1 \text{ pixel} \cdot \text{mm}^{-1}$.

The distribution of PF types in this study is categorized on the basis of the classification criteria proposed by Weiler and Flühler (2004) and further subdivided according to the morphology of dye tracer patterns and types of subsurface flow, as outlined by Li et al. (2018a,b), as shown in Table S1.

2.6. Meta-analysis

The literature was collected up to December 2023 from the Web of Science and China National Knowledge Infrastructure (CNKI). The search terms included “preferential flow,” “dye tracing,” or “Brilliant Blue staining,” “vegetation restoration,” or “land use change,” and “Loess Plateau.” The selection criteria for relevant papers were as follows: (1) the inclusion of at least one land use or vegetation type (e.g.,

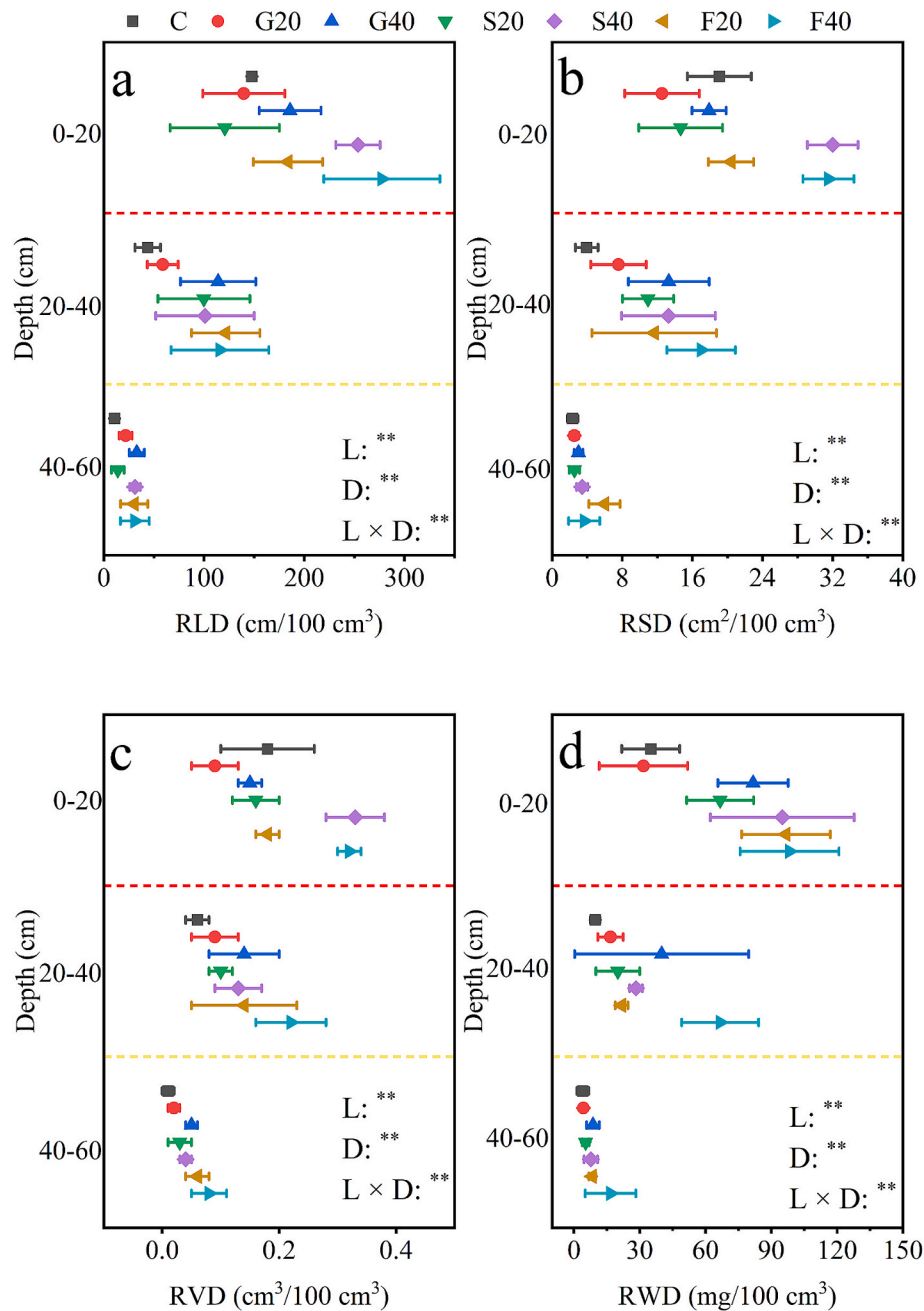


Fig. 6. Root characteristics at different depths and land-use types. D: depth, L: land-use type, $\times$: interaction, ** $P < 0.01$.

forestland, shrubland, grassland, or bare land), (2) the conversion of all current land use types from farmland, (3) clear indication of repeated experiments, and (4) the use of the constant head method and Brilliant Blue dye tracing for measuring infiltration and PF indicators. Owing to variations in soil, vegetation, climate, and other factors (Li et al., 1985), the LP was divided into three ecological zones (EZ-I, EZ-II, and EZ-III) to examine the distribution characteristics of PF across these zones. In total, 139 observations from 13 publications were collected for the LP of China. The list of references is in Table S2.

2.7. Statistical analysis

One-way analysis of variance (ANOVA) was employed to examine differences in infiltration properties and PF indicators (e.g., MDD, DC, MFD, PF_i , and LI) across different land-use types. Moreover, ANOVA was used to examine the differences in soil physicochemical properties and

root characteristics among different land use types at the same soil depth. Two-way ANOVA was used to assess the effects of land-use type and soil depth on the soil properties and root characteristics. PCA was used to calculate the PF-CI. Before computation, different PF indicators were normalized to eliminate the influence of dimensional (unit) differences and ensure balanced variable contributions. Redundancy analysis (RDA) and variance partitioning analysis (VPA) were used to quantify the relative contributions of soil physical and chemical properties and plant roots to the PF-CI (Rousseau et al., 2006; Liao et al., 2023). Partial least squares path modeling (PLS-PM) was used to examine the causal relationships between PF-CI and its significant influencing factors (e.g., SOM, AGC, and MP), with model fit assessed on the basis of goodness of fit (Hwang & Cho, 2020). One-way and two-way ANOVA were performed using SPSS Statistics ver. 26.0 (SPSS Inc., Chicago, IL, USA). PCA, RDA, VPA, and PLS-PM were conducted using R version 4.0.3. The detailed data and processing methods can be found in

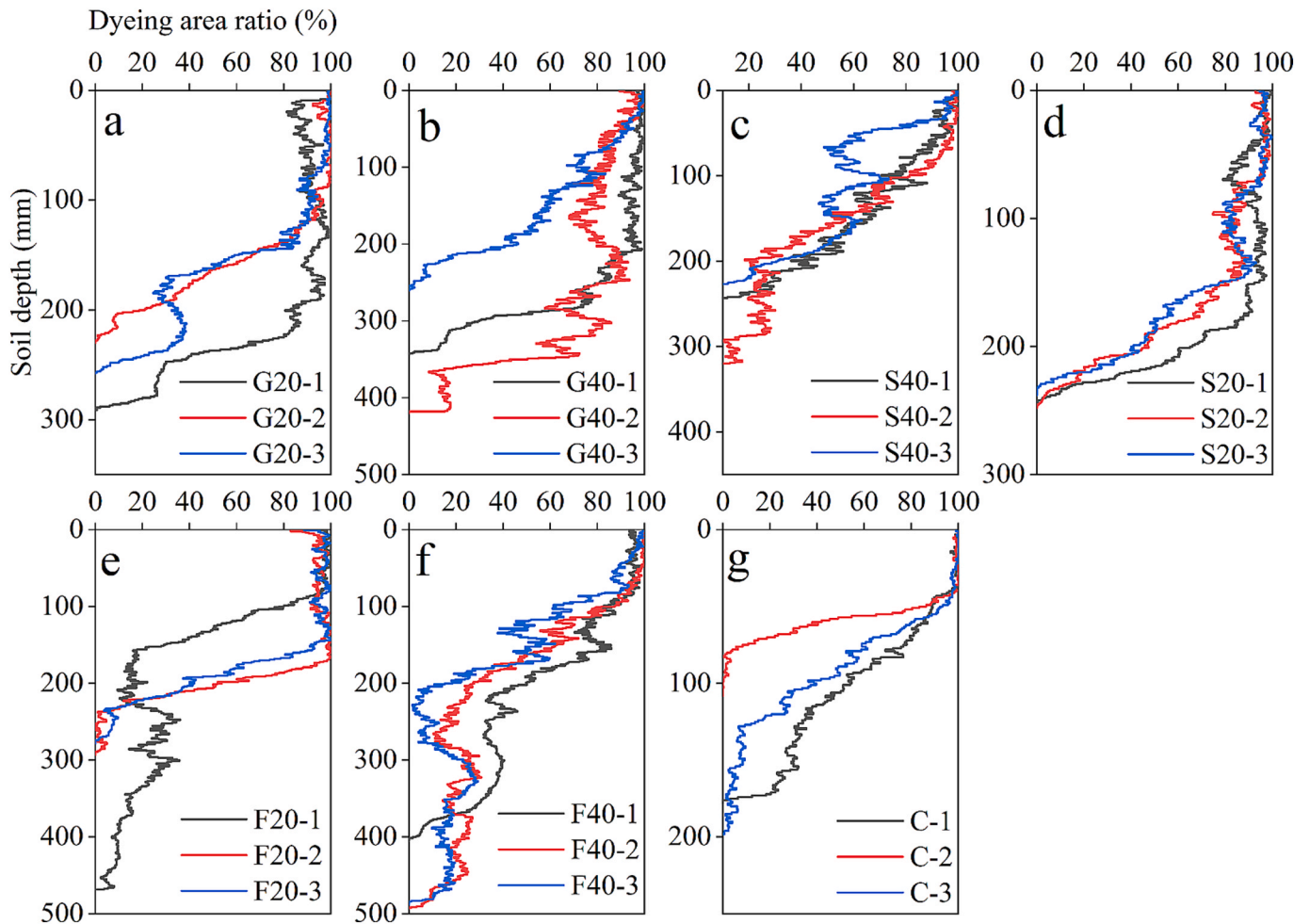


Fig. 7. Variation in the dyed area ratio with soil depth for different land-use types and vegetation restoration years, -1, -2, and -3 represent the three test points.

Table 4

The degree index of the PF in the vertical section of different land-use types.

Land-use	MDD (cm)	DC (%)	MFD (cm)	PF _f (%)	LI (%)	CV
F20	34.30(8.80)b	58.68(14.72)bcd	19.16(4.23)bc	40.36(2.82)ab	329.67(29.48)b	1.26(0.09)bc
F40	45.85(4.06)a	45.75(8.62)d	27.23(1.93)a	43.88(3.64)a	386.00(8.73)a	0.98(0.06)c
S40	29.14(5.89)b	55.41(6.30)bcd	16.59(0.74)bc	41.5(8.32)ab	337.67(30.61)b	1.31(0.08)bc
S20	24.02(0.64)bc	70.54(1.94) ab	15.99(0.76)c	33.45(2.33)bc	242.77(6.38)c	1.35(0.03)bc
G20	25.81(2.57)bc	68.57(2.86)abc	16.23(2.60)bc	36.11(3.19)abc	224.41(9.35)cd	1.40(0.05)b
G40	33.96(6.45)b	52.87(2.73)cs	22.38(5.29)ab	42.07(0.12)ab	320.12(10.51)b	1.04(0.31)bc
C	18.04(1.23)c	75.39(3.06)a	8.00(2.26)d	29.43(1.98)c	187.78(26.65)d	2.20(0.43)a
Sig. of ANOVA	0.002	0.011	<0.01	0.025	<0.01	<0.01

The values outside the brackets represent the mean, while the values inside the brackets indicate the standard deviation. Different letters in the same column indicate a significant difference at the $P < 0.05$ level.

Fig. 4.

3. Results

3.1. Changes in soil physicochemical properties after vegetation restoration

The soil physicochemical properties were significantly influenced by the soil depth, land-use type, and year of vegetation restoration ($P < 0.05$). Additionally, the interaction between depth and land-use type had a statistically significant effect on soil clay, silt, sand, MP, Con, and FD ($P < 0.05$) (Table 3). Across the different soil layers, the BD in the cropland was consistently greater than that in the revegetation land,

indicating that the retirement of land significantly reduced the BD. In terms of texture, the clay content in cropland was generally lower than that in revegetation land, especially in the surface soil, where it accounted for only 10.05 %. Conversely, the sand content in croplands was greater than that in revegetation lands (except for G20), with a maximum of 26.25 %, which was also observed in the surface layer. Compared with that in cropland, the SOM in revegetation land was significantly greater (except for G20 and F20), with a more pronounced increase in G40. Forestland, shrubland, and grassland generally increased the pH value in the 0–60 cm depth range, although only G40 showed statistically significant differences in the 0–20 cm and 20–40 cm depth ranges ($P < 0.05$). Similarly, the AGC in these revegetation lands increased overall in the 0–60 cm depth range, but statistically significant

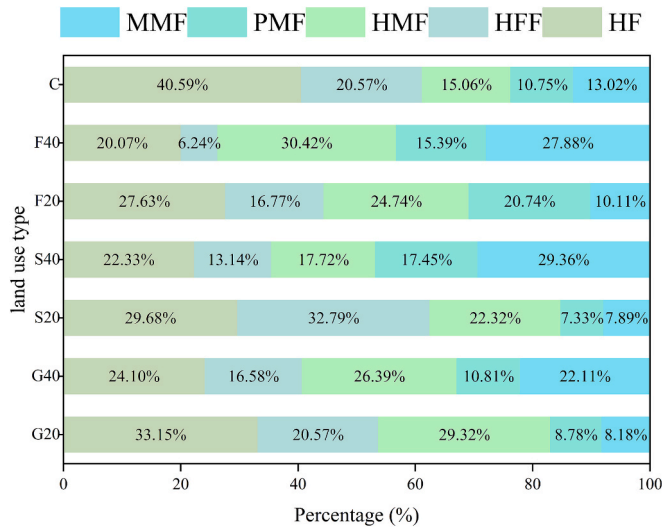


Fig. 8. Distribution characteristics of different flow types under different land use types. MMF: Mixed macropore flow, PMF: Poorly interacted macropore flow, HMF: Highly interacted macropore flow, HFF: Heterogeneous figure flow, HF: Homogeneous flow.

differences were observed only in F40 in the 0–20 cm depth range ($P < 0.05$). Compared with those in cropland, the MP, Con, and FD in revegetation land were significantly greater and increased with increasing vegetation restoration duration (Table 3). As shown in Fig. 5, the MP values were significantly greater in the revegetation plots than in the cropland plots across the 0–20 cm, 20–40 cm, and 40–60 cm soil layers ($P < 0.05$). Moreover, MP gradually decreased with increasing soil depth, and its variability also decreased.

3.2. Changes in root characteristics after vegetation restoration

Vegetation restoration promotes the growth and development of plant roots. The RLD, RSD, RVD, and RWD of the revegetation land (except for G20 and S20 at 0–20 cm) were significantly greater than those of the cropland ($P < 0.01$). These root metrics decreased with increasing soil depth but increased with increasing duration of vegetation restoration. The interaction effect between soil depth and land-use type also had a statistically significant effect on RLD, RSD, RVD, and RWD ($P < 0.01$). At the 0–20 cm soil depth, S40 presented the highest RSD and RVD, with values of $32.00 \text{ cm}^2/100 \text{ cm}^3$ and $0.33 \text{ cm}^3/100 \text{ cm}^3$, respectively. F40 had the highest RLD and RWD at the same depth, with values of $57.82 \text{ cm}/100 \text{ cm}^3$ and $98.33 \text{ mg}/100 \text{ cm}^3$, respectively (Fig. 6). The minimum RLD, RSD, RVD, and RWD occurred in G20 and the 20-year shrubland (S20). With increasing soil depth, the RLD, RSD, RVD, and RWD of the cropland decreased. At 20–40 cm, the RLD, RSD, RVD, and RWD of the cropland decreased to $43.80 \text{ cm}/100 \text{ cm}^3$, $3.95 \text{ cm}^2/100 \text{ cm}^3$, $0.06 \text{ cm}^3/100 \text{ cm}^3$, and $9.67 \text{ mg}/100 \text{ cm}^3$. At 40–60 cm, the RLD, RSD, RVD, and RWD of the cropland decreased to $10.72 \text{ cm}/100 \text{ cm}^3$, $2.39 \text{ cm}^2/100 \text{ cm}^3$, $0.01 \text{ cm}^3/100 \text{ cm}^3$, and $4.00 \text{ mg}/100 \text{ cm}^3$, respectively. The root indices (RLD, RSD, RVD, and RWD) of F20 and F40 were significantly greater than those of the other land-use types. In summary, vegetation restoration significantly promoted root morphological development and increased plant biomass.

3.3. Changes in PF after vegetation restoration

Significant variations were observed in the dyed area ratio along the vertical profile of the soil. Compared with that of cropland, the MDD significantly increased in revegetation land (Fig. 7). As shown in Table 4, the MDD values of F20, F40, S20, S40, G20, and G40 were 34.30 cm, 45.85 cm, 24.02 cm, 29.14 cm, 25.81 cm, and 33.96 cm, respectively,

which were significantly greater than those of the cropland (18.04 cm). The MFD in cropland was 8.00 cm, which was lower than those in F20 (19.16 cm), F40 (27.23 cm), S20 (15.99 cm), S40 (16.56 cm), G20 (16.23 cm), and G40 (22.38 cm). The PF_f results indicated that only 29.43 % of the dyed area in cropland corresponded to PF paths. In contrast, these values were 40.36 %, 43.88 %, 33.45 %, 41.50 %, 36.11 %, and 42.07 % for F20, F40, S20, S40, G20, and G40, respectively. The LI for cropland (187.78) was lower than those of F20 (329.67), F40 (386.00), S20 (242.77), S40 (337.67), G20 (224.41), and G40 (320.12), indicating that the PF paths in revegetation land were more irregular than those in cropland.

As shown in Fig. 8, there were significant differences in the proportions of water flow types in different land-use types. Overall, the proportion of HF in croplands was greater than that in revegetation lands, whereas the proportion of PF was lower. The proportion of PF also increased with the duration of vegetation restoration. The proportion of HF in cropland reached 40.59 %, whereas in revegetation lands, it ranged from 20.07 % to 33.15 %. Among the revegetation lands, G20 had the highest proportion, whereas F40 had the lowest proportion. The HMF proportion in cropland was only 15.06 %, whereas in revegetation lands, it ranged from 17.72 % to 30.42 %, with the highest occurrence observed in F40. These findings indicate that vegetation restoration enhances the development of PF paths. Fig. 9 illustrates that after farmland was converted to revegetation land, the PF indices changed. Specifically, MDD, MFD, and LI increased, whereas DC decreased. The PF paths became deeper, received more flow, and contributed more to infiltration (Fig. 9a). The PCA results revealed that the first axis accounted for 79.9 % of the variance in the staining pattern for these parameters, indicating that PF-CI is a suitable comprehensive index for evaluating PF (Fig. 9c). Among the indicators, the PF_f and MDD contribute more than 20 % to the PF-CI (Fig. 9b). The PF-CIs for cropland, G20, G40, S20, S40, F20, and F40 were -3.00 , -1.36 , 1.15 , -1.33 , -0.61 , 2.23 , and 3.10 , respectively (Fig. 9d). The PF-CI increased from -3.00 to 3.10 , further confirming that the conversion from cropland to forestland, shrubland, and grassland promoted the development of PF.

The infiltration rate initially decreased rapidly over time and then gradually stabilized after 20 min (Fig. 10). Across different land-use types, the infiltration rates of PF were consistently higher than those of MF (Fig. 10a–g). Under the MF infiltration mode, no significant differences were detected between cropland and revegetation land in terms of IIR and SIR (Fig. 10h and i). However, under the PF infiltration mode, the IIR-P values for cropland, G20, G40, S20, S40, F20, and F40 were 0.42, 1.08, 1.15, 0.98, 1.13, 1.27, and 1.49, respectively, whereas the SIR-P values were 0.07, 0.21, 0.22, 0.15, 0.18, 0.20, and 0.28, respectively. The IIR-P and SIR-P values for revegetation land were significantly greater than those for cropland ($P < 0.01$), with the highest infiltration rates observed in F40 (Fig. 10h and i). Significant differences ($P < 0.01$) were detected between the IIR values of MF and PF across all land-use types. However, significant differences between SIR-M and SIR-P were detected only in F40 ($P < 0.01$), whereas no significant differences were detected among the other revegetation lands.

3.4. Relationship between PF and soil physicochemical properties and root characteristics

Pearson correlation analysis revealed that IIR-M was significantly positively correlated with SOM, AGC, MP, and Con ($P < 0.05$). In contrast, IIR-F was significantly negatively correlated with BD, clay, silt, and IMC ($P < 0.05$) and significantly positively correlated with sand, AGC, MP, Con, FD, RWD, RLD, RSD, and RVD ($P < 0.01$). The SIR-M was significantly positively correlated with AGC, MP, and Con ($P < 0.05$). The SIR-F was significantly negatively correlated with BD, clay, and silt ($P < 0.05$) and significantly positively correlated with sand, AGC, MP, Con, FD, RWD, RLD, RSD, and RVD ($P < 0.05$). PF-CI was significantly positively correlated with SOM, sand, MP, Con, FD, RWD, RLD, RSD, and RVD ($P < 0.05$), whereas it was significantly negatively correlated with

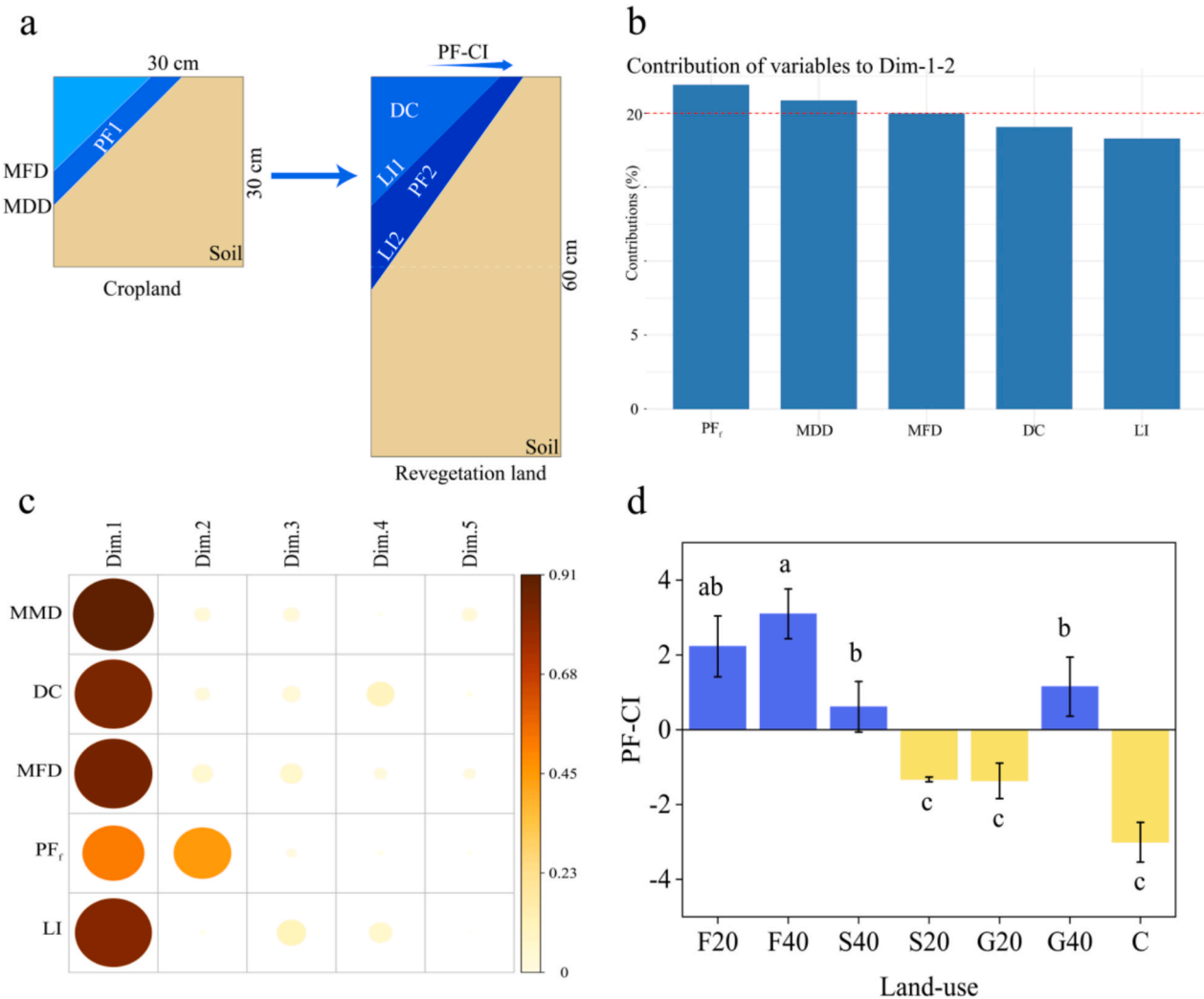


Fig. 9. a Represents the change process of the PF index from cropland to restoration land DC: the area ratio of triangle to rectangle, MFD: vertical length of light blue triangle, MDD: vertical length of dark blue triangle, LI1: the length index of cropland, denoted by the bevel of a right triangle, LI2: the length index of restoration land, denoted by the bevel of a right triangle, PF1: preferential flow of cropland, PF2: preferential flow of reatoration land. b represents the contribution of each PF indicator to the first and second principal component axes in the PCA, c represents the representativeness of each principal component axis for all PF indicators, d represents the change in the PF-CI under different land-use types and vegetation restoration years, and the lowercase letters in the Fig. 9d indicate significant differences in the PF-CI among different land use types at the 0.05 level.

IMC ($P < 0.05$) (Fig. 11). RDA indicated that RDA1 and RDA2 explained 72.66 % and 13.7 % of the variance in response variables, respectively (Fig. 12a), and the results of the Monte Carlo permutation test were consistent with those of the Pearson correlation analysis (Table S3). The soil physical properties, chemical properties, and root characteristics explained 61.2 %, 23.2 %, and 73.1 %, respectively, of the variance in the response variables, with 15.6 % of the variance remaining unexplained. The combined explanatory power of root characteristics and soil physical properties was 39.7 %, indicating that these are the primary factors influencing PF. The individual explanatory rates for root characteristics, soil physical properties, and chemical properties were 23.1 %, 4.9 %, and 3.3 %, respectively, highlighting that root characteristics are the dominant factors affecting PF (Fig. 12b).

4. Discussion

4.1. Effects of vegetation restoration on soil properties and root characteristics

Significant vegetation restoration or land-use conversion has profoundly altered the soil structure and subsurface root systems. Extensive

research has confirmed that vegetation restoration can reduce BD and increase MP (Qiu et al., 2022; Liao et al., 2023). Our study revealed that cropland presented the highest BD and the lowest MP, primarily because of compaction from tillage and other farming activities. In contrast, revegetation lands generally have a lower BD and higher MP, which is a result of reduced anthropogenic disturbance and the prolonged influence of plant roots and soil fauna. Additionally, as the duration of vegetation restoration increased, the MP, Con, and FD of the soil pores gradually improved (Table 3). In this study, CT scanning was employed to characterize the MP and to quantify the FD and Con of these features. MP, FD, and Con exhibited significant variability, with the characteristics of the macropores in the revegetation land being markedly greater than those in the cropland and further improving with increasing number of years of revegetation (Fig. 5). This conclusion aligns with findings from studies on natural vegetation restoration on the LP (Zheng et al., 2023).

Compared with that in cropland, the relative proportion of sand in the soil decreased after vegetation restoration, whereas the contents of silt and clay increased. This change is primarily due to the increased erosion resistance provided by vegetation, which intercepts smaller soil particles, such as clay and silt (Wang et al., 2024). Our study also

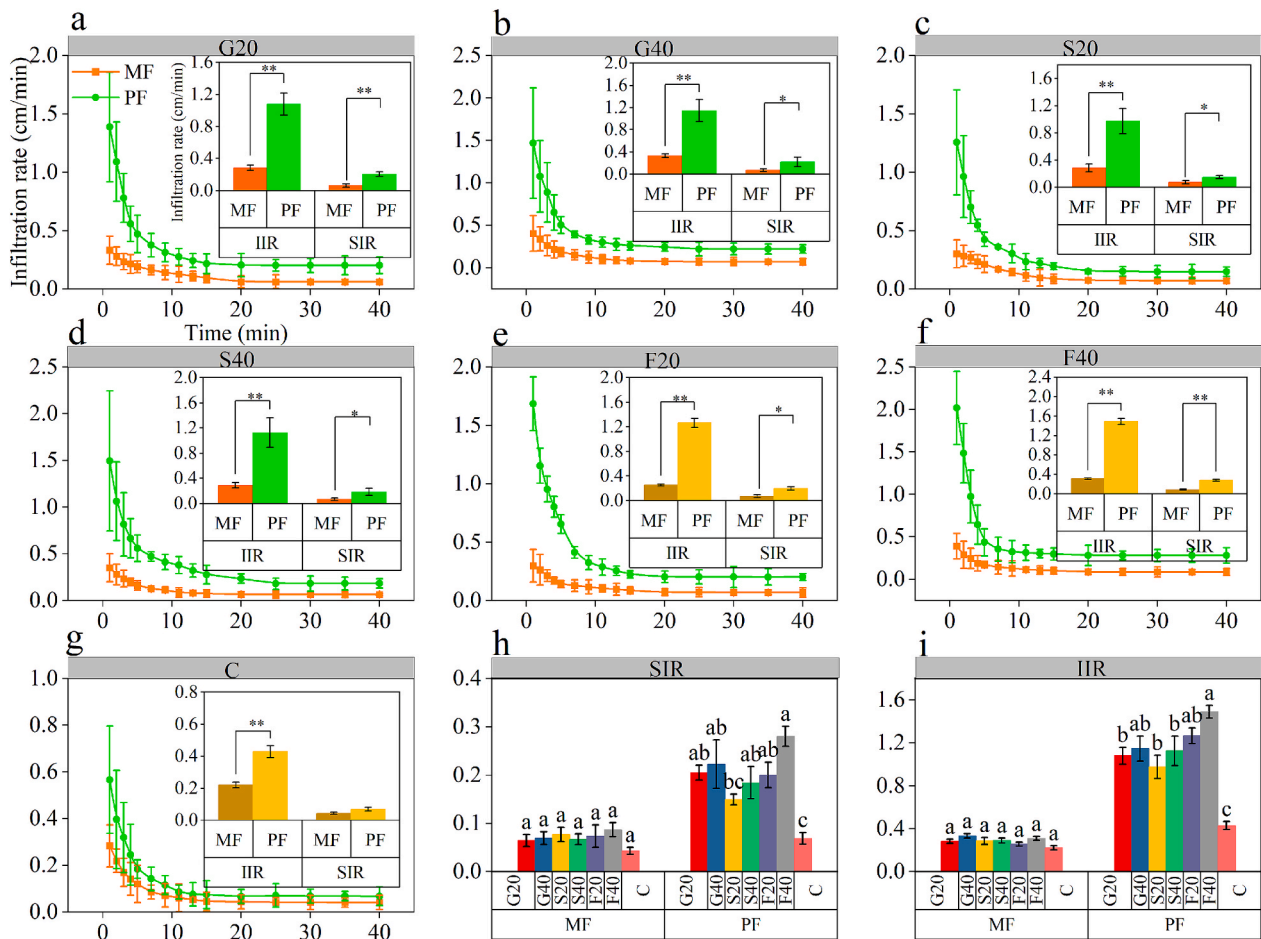


Fig. 10. Changes in PF and MF infiltration rates with infiltration time, * indicates the difference between the PF infiltration rate and the MF infiltration rate under IIR or SIR, according to the *t* test, ** $P < 0.01$, * $P < 0.05$, and the lowercase letters indicate the difference between different land-use types at the level of $P = 0.05$.

revealed that vegetation restoration generally increases the SOM (except for G20 and F20) and AGC (Table 3). This is attributed mainly to the decomposition of tree litter, shrub leaves, and the abundant grass roots in grasslands, which, under the influence of soil fauna and microorganisms, transform into rich organic matter and soil aggregates. Additionally, the thicker litter layer in restored areas reduces soil erosion and the associated loss of organic matter, whereas crop residues in croplands are more likely to be removed by human activities (Beven & Germann, 2013; Sun et al., 2018). Our study revealed that the pH in revegetation land is greater than that in cropland, which contrasts with the findings of Qiu et al. (2022) and Gu et al. (2019) (Table 3). The primary reason for this difference is that our cropland soil samples were collected before artificial fertilization, resulting in lower soil salinity. After fertilization, nutrients such as carbon and nitrogen in the cropland soil are absorbed by crops, leading to a significant accumulation of salts in the soil, which increases the pH.

Plant root systems play crucial roles in influencing soil structure and water dynamics, thereby affecting the development of PF (Luo et al., 2019a; Wu et al., 2021). Our study demonstrated that, except for G20 and S20 at 0–20 cm, most revegetation lands presented higher RLD, RSD, RVD, and RWD values than did croplands (Fig. 6). In all the plots, the roots were primarily concentrated from 0–20 cm, with the root traits gradually decreasing with increasing soil depth. Among the different revegetation lands, forestland presented higher root trait parameters, particularly in deeper soil layers, and these parameters increased as forest age increased. This is due mainly to the robust root systems of trees, which, as they age, generate more decayed roots, thereby creating more macropore networks that serve as preferential pathways for water

flow (Cai et al., 2024). Studies have indicated that decayed roots may form larger macropore networks more readily than living roots do (Kiepe, 1995; Chisci et al., 2001). These findings are consistent with the results of Tian et al. (2023) and Cai et al. (2024) but contrast with those of Luo et al. (2019b), who argued that large roots contribute less to preferential water transport than fine roots do, as fine roots constitute a greater proportion of the total root mass in the soil profile.

4.2. Effects of vegetation restoration on PF

The meta-analysis results indicate that the MDD in revegetation land was generally greater than that in farmland, whereas the DC was lower (except for shrubland in EZ-I). In EZ-I, shrubland presented the greatest median MDD (46.04 cm), which was significantly greater than that in farmland (27.84 cm) ($P < 0.01$), and had a larger DC than farmland. In EZ-II, although there were no significant differences in the MDD and DC between revegetation land and farmland, shrubland and grassland presented greater MDDs than farmland did, though forestland did not. Conversely, the DC of forestland, shrubland, and grassland was lower than that of farmland, with shrubland showing the highest median MDD (41.91 cm) and grassland the lowest median DC (38.65 %). In EZ-III, the MDD significantly decreased from forestland to shrubland to grassland, while DC progressively increased. After vegetation restoration on the LP, changes in MDD and DC varied across different ecological zones. However, overall, the MDD in revegetation land was greater than that in farmlands, whereas the DC was smaller (Fig. 13). This finding aligns with our experimental results (Fig. 7).

Vegetation restoration increases the soil infiltration capacity by

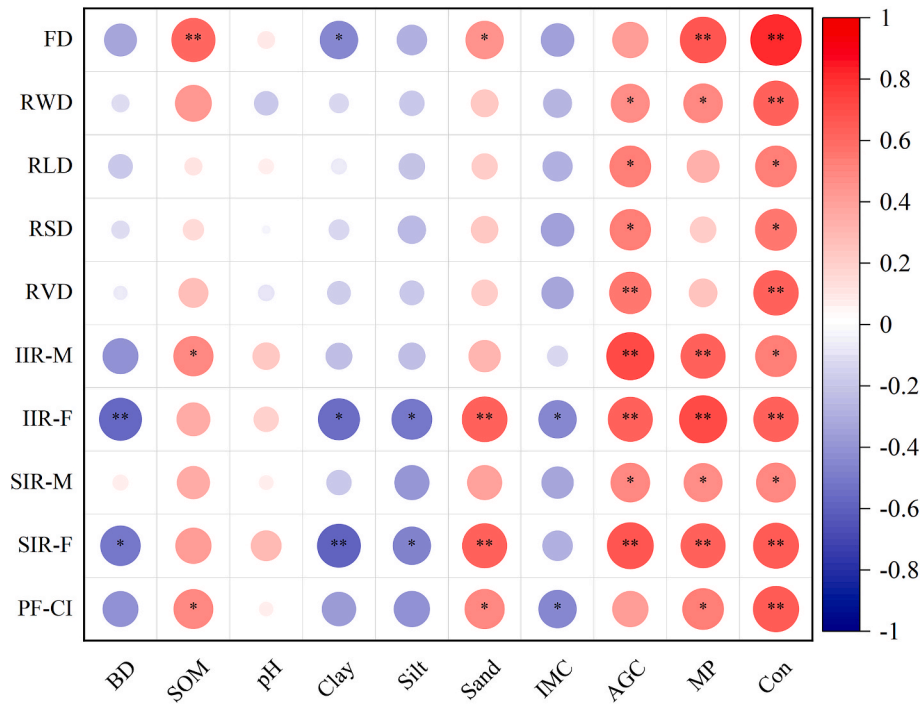


Fig. 11. The Pearson correlation matrix between the PF index and each influencing factor shows that red indicates a positive correlation, blue indicates a negative correlation, and the size of the circle represents the correlation strength. * indicates a significant difference at the 0.05 level, ** indicates a significant difference at the 0.01 level.

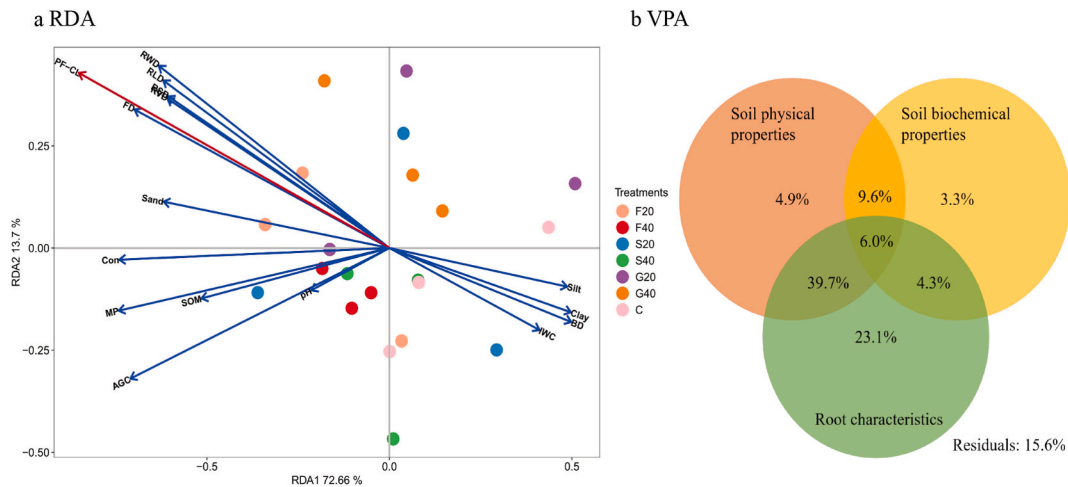


Fig. 12. a Shows the RDA, where the soil physicochemical properties and root characteristic parameters are represented by blue arrows, and the PF-CI is indicated by red arrows. The values on the RDA axes reflect the percentage of total variation explained by each axis. b depicts the VPA, with soil physical properties including BD, clay, silt, sand, IMC, MP, Con, and FD; soil chemical properties including SOM, pH, and AGC; and root characteristic parameters such as RLD, RSD, RVD, and RWD. The percentage values represent the contributions of soil physical and chemical properties, and plant root characteristics to the variation in PF-CI.

promoting the development of PF (Fig. 10). This may be attributed to vegetation restoration altering the soil structure, which shortens the response time between soil moisture changes and accelerates the downwards movement of the wetting front. Consequently, this enhances PF development and ultimately improves the soil infiltration capacity (Ge et al., 2022). Moreover, the longer the vegetation restoration duration is, the more strongly PF develops. (Table 4 and Fig. 7). These findings are also supported by the studies of Wang et al. (2020) and Cai et al. (2024). The impact of vegetation restoration on PF has also shown similar patterns in other regions. Compared with cropland, natural forestland presented greater MDD, smaller DC, higher PF_f, and greater LI in the karst regions of China (Ding et al., 2023). Compared with

farmland, forestland and grassland significantly increased the soil infiltration capacity and intensity of PF in the Morogoro region of Tanzania (Lulandala et al., 2022). Collectively, these results demonstrate that vegetation restoration enhances the development of PF.

4.3. Effects of soil and roots on PF

Many studies have demonstrated that soil physicochemical properties and plant root systems are key factors influencing PF (Zhang et al., 2017b). Our findings revealed that PF-CI was significantly positively correlated with SOM, sand, MP, Con, FD, RWD, RLD, RSD, and RVD ($P < 0.05$) and negatively correlated with IMC ($P < 0.05$) (Fig. 11). To further

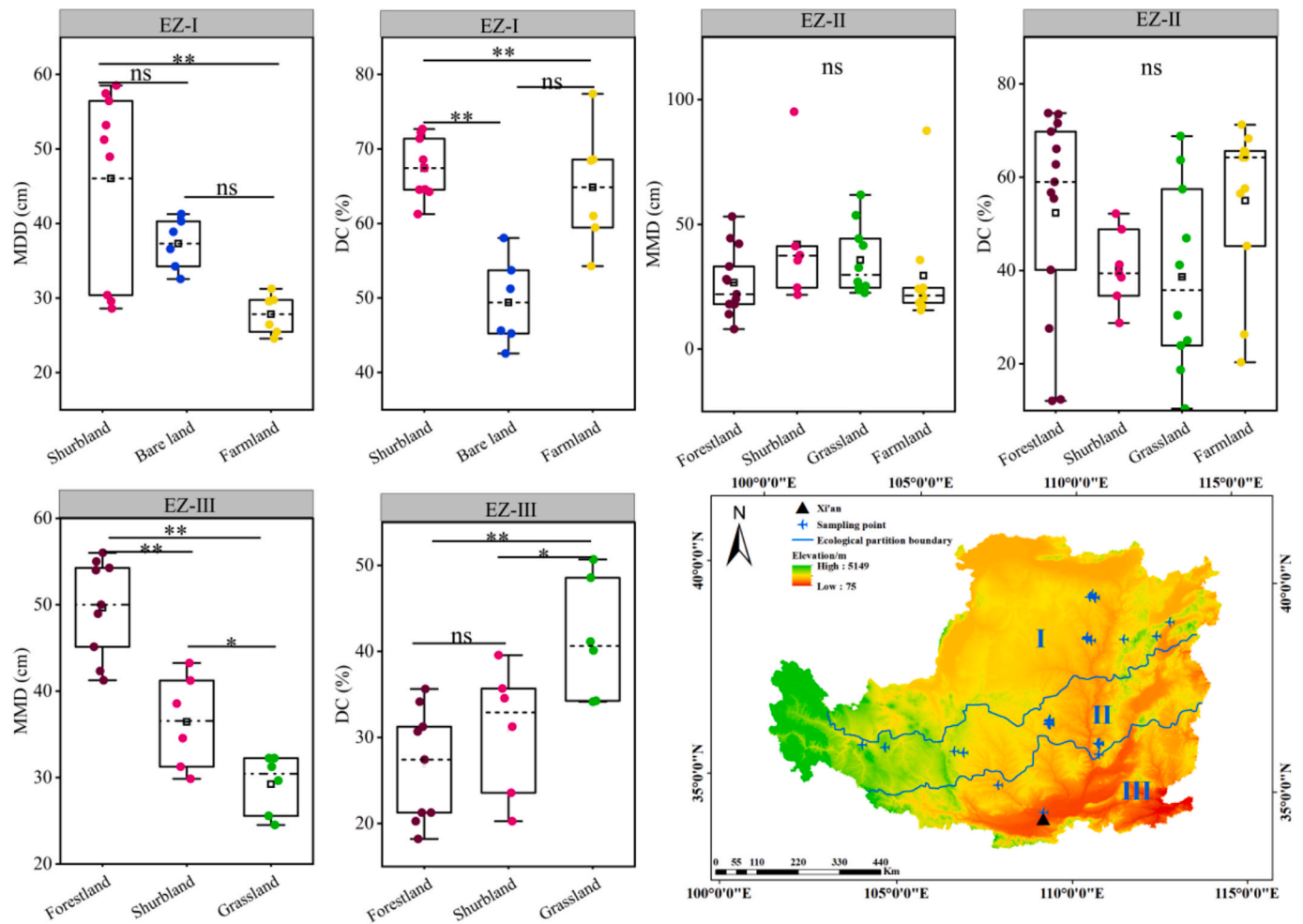


Fig. 13. Characteristics of the MDD and DC across different land-use types on the Loess Plateau. The boxes represent the interquartile range, with the line inside each box indicating the median value. The whiskers extend to the 5th and 95th percentiles. * indicates significance at $P < 0.05$, ** indicates significance at $P < 0.01$, and ns denotes not significant ($P > 0.05$).

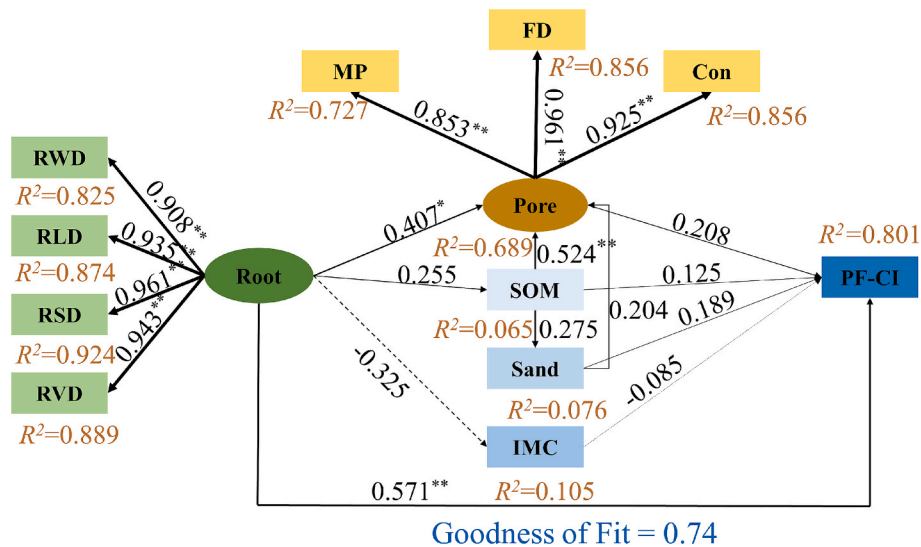


Fig. 14. The partial least squares path modeling (PLS-PM) illustrates the effects of soil physical and chemical properties, as well as root characteristic parameters on PF-CI. The width of the lines reflects the strength of the causal relationships, corresponding to the standardized path coefficients indicated near each line. R^2 denotes the proportion of variance in the dependent variable explained by the model. Coefficients marked with * and ** are significant at the 0.05 and 0.01 probability levels, respectively. The Goodness of Fit statistics are displayed below the modelling framework.

quantify the effects of soil and root parameters on PF, we developed a partial least squares path modeling. The model results indicate that the model explains 80.1 % of the variance in the PF-CI. Compared with soil physicochemical properties, plant roots had the strongest and most direct impact on PF ($\beta = 0.571$, $P < 0.01$). Among the soil physicochemical properties, the soil macropore structure had the strongest influence on PF ($\beta = 0.208$), followed by sand ($\beta = 0.189$), SOM ($\beta = 0.125$), and IMC ($\beta = -0.085$). Moreover, the soil macropore structure was significantly affected by plant roots ($\beta = 0.407$, $P < 0.01$) (Fig. 14). In summary, the macropore structure (MP, Con, and FD) and plant root parameters (RLD, RSD, RVD, and RWD) are the primary factors influencing the development of PF (Cai et al., 2024).

Soil macropores serve as the primary conduits for PF, playing a crucial role in soil water movement and the preferential transport of surface pollutants (Kang et al., 2022; Li et al., 2023). Vegetation restoration on the Chinese LP has significantly improved soil macropore structures—enhancing MP, and FD—thereby increasing PF (Zheng et al., 2023; Zhao et al., 2023). Li et al. (2018a,b), via CT scanning, reported that with the restoration of vegetation on the LP, the volume and number of macropores in forestland and grassland soils increased significantly, especially in the 0–20 cm surface layer, which is consistent with our findings (Fig. 5). Ma et al. (2022) demonstrated that soil columns with lower MP but higher Con exhibit much greater K_s than those with greater MP but poorer Con. Therefore, theoretically, Con has a greater impact on soil water movement than does MP alone, suggesting that comparing MP in isolation does not directly reflect the infiltration capacity of different soils. Jiang et al. (2019) reported that more live roots, decayed roots, and termite mounds were observed in forestland, shrubland, and grassland than in farmland. This finding explains why PF was significantly positively correlated with MP and SOM. Furthermore, the results indicate that among the soil physicochemical properties, MP is the greatest contributor to the variation in PF, which aligns with the findings of Flury et al. (1994).

Compared with soil MP, the morphological characteristics of plant roots have a more significant effect on PF (Fig. 12), which is consistent with the findings of Perillo et al. (1999), who identified both decayed and live root systems as critical factors influencing PF pathways. Similarly, Cui et al. (2022) emphasized the key role of fine roots in affecting the soil infiltration potential in semiarid regions. During root growth, plant roots not only extend into PF channels but also create pathways through biopores, cracks, and cavities (Cai et al., 2024). Our study revealed that root growth was associated with increased MP, Con, FD, and SOM, and decreased sand content (Table 3 and Fig. 6). This is supported by Benegas et al. (2014), who highlighted the importance of root input in enhancing soil properties during vegetation restoration, thus promoting PF. The root characteristic parameters were positively correlated with the PF-CI in our study (Fig. 11 and Fig. 12). The distribution of stained areas matched the root distribution, and PF was absent in layers with fewer roots. In areas with longer vegetation restoration durations, higher values of RWD, RVD, RSD, and RLD were observed than in sites with shorter restoration durations (Fig. 6). Consequently, the PF in F40, S40, and G40 was significantly greater than that in F20, S20, and G20, which aligns with the findings of Zhang et al. (2017b). In the soil profile, isolated stained patches are often associated with coarse roots (Fig. S2), corroborating the findings of Devitt and Smith (2002), who noted that coarse roots present greater hydraulic conductivity than do fine roots. However, Cui et al. (2022) reported that coarse roots, which decay more slowly, can compact the soil and impede water flow, thereby increasing RWD and reducing soil infiltration. This discrepancy may arise from differences in root morphology and distribution across plant species.

5. Conclusion

In this study, we quantitatively analysed the changes in PF characteristics under different vegetation types and years of restoration on the

LP, as well as their responses to soil physicochemical properties and root characteristics. Vegetation restoration modifies soil physicochemical properties and root characteristics, thereby influencing the development of preferential flow. Revegetation soils exhibit abundant and well-connected macropores, higher aggregate stability, and extensive root systems, all of which enhance preferential flow pathways. The extent of preferential flow intensifies with increasing vegetation restoration duration but decreases with increasing soil depth. Preferential flow contributes to more than 50 % of the total infiltration, with its infiltration rate consistently exceeding that of matrix flow. The soil pore structure and root characteristics are the primary drivers of preferential flow, jointly explaining 39.7 % of its variability. Plant root systems facilitate preferential flow both directly, by forming interconnected preferential flow pathways, and indirectly, by enhancing soil pore structure and connectivity, thereby promoting the development of preferential flow. These results are important for understanding the hydrological effects of soil on vegetation restoration on the LP and other arid and semiarid regions.

CRediT authorship contribution statement

Yongfu Wen: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Mengzhen Li:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Peng Gao:** Writing – review & editing, Supervision, Funding acquisition. **Jiahui Zhou:** Writing – review & editing. **Xucheng Ai:** Writing – review & editing, Data curation. **Yuxuan Zhang:** Writing – original draft, Data curation. **Xingmin Mu:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2025.133349>.

Data availability

Data will be made available on request.

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